

Niyar R BARMAN

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RESEARCH INTERESTS

Efficient LLMs · Model Compression · Knowledge Distillation · Mechanistic Interpretability

Understanding how capabilities are represented within model parameters, how those representations change as capacity is reduced, and what is preserved or lost during pruning and distillation. I am particularly interested in identifying the structures that support mathematical and multi-step reasoning, and using that understanding to develop compression methods that preserve those computations and help models reason better in substantially smaller architectures

EDUCATION

Université de Toulouse, France

Master 2 Interactions of Computer Science and Mathematics for AI 2025–2026

National Institute of Technology Silchar, India

B.Tech. Electronics and Communication Engineering 2021–2025

EXPERIENCE

LINAGORA Labs, France

Research and Development Intern (Full Time) Apr 2026–Sep 2026

- Investigated how structured pruning and knowledge distillation preserve capability in large language models, developing an end-to-end compression pipeline for the 23B-parameter Luciole model and producing 14B and 8B students at a 64% parameter reduction on a Slurm-managed NVIDIA H100 cluster using PyTorch, Megatron-Core, Megatron-Bridge, and NVIDIA TensorRT Model Optimizer.
- Trained across 274B tokens and roughly 30,500 H100-hours on 128 GPUs with tensor, pipeline, context, and data parallelism at 32K sequence length, running 50B-token FP8 continued pretraining and two 112B-token logit-distillation runs.
- Delivered an 8B student that outperforms the size-matched production baseline on 9/9 English and French math benchmarks, raising mean math accuracy from 31.7 to 49.8 while retaining 99.2% of the 23B teacher's math ability at 36% of its parameters; the 14B student exceeded the original 23B base on 9/9.
- Rebuilt a 179B-token corpus into a 112B-token cleaned training pool through global exact deduplication and document-length capping, cutting packed training windows containing no end-of-sequence signal from 51.8% to 7.3% and resolving a generation-collapse failure that had blocked an earlier training run.

Institut de Recherche en Informatique de Toulouse (IRIT), France

Research Intern (Full-time) May–Jul 2024; Jan–Mar 2025; Sep 2025–Mar 2026

- Researched discourse reasoning and numerical comprehension in large language models under Dr. Nicholas Asher
- Designed and optimized training pipelines for 125M parameter models on multi-GPU HPC clusters, reducing training time through efficient memory management and parallel processing
- Implemented multi-GPU optimization pipeline (up to 12 GPUs) for task-aware layer elimination across 4 LLM families (7B–13B parameters), achieving up to 60% reduction in inference TFLOPs while improving accuracy on 9 benchmarks
- Implemented reproducible experiments using PyTorch, DDP, and automated evaluation pipelines
- Developed a custom fused Triton CUDA kernel replacing softmax attention with Scaled Signed Averaging (SSA), a learnable parameterized score transformation with corrected gradient derivations and log-exp reformulation for numerical stability, achieving 3x speedup over baseline PyTorch attention and matching Megatron-optimized softmax performance across 30K+ training steps on a 114M parameter Nemotron model
- Trained a SSA attention 1.3B parameter Nemotron-3 architecture model from scratch on a 64xH200 GPU cluster (CALMIP KAIROS)

Artificial Intelligence Institute, University of South Carolina

Research Intern (Remote) Jan 2023–Dec 2024

- Built automated evaluation pipelines for systematic analysis of 15 large-scale models, processing millions of samples with statistical rigor
- Designed pipelines for statistical testing and task-specific evaluation
- Conducted large-scale experiments on visual paraphrasing and watermarking robustness, highlighting vulnerabilities of current methods

PUBLICATIONS

Counter Turing Test CT^2 : AI-Generated Text Detection is Not as Easy as You May Think – Introducing AI Detectability Index

M. Chakraborty, S. M. T. I. Tonmoy, S. M. M. Zaman, K. Sharma, N. Barman, C. Gupta, S. Gautam, T. Kumar, V. Jain, A. Chadha, A. P. Sheth, and A. Das

EMNLP 2023 | Outstanding Paper Award

TELL-TALE: Task Efficient LLMs with Task Aware Layer Elimination

O. Naim, K. Sharma, N. Barman, and N. Asher

ACL 2026 Findings

DIMSUM: Discourse in Mathematical Reasoning as a Supervision Module

N. Barman, K. Sharma*, N. Asher, and A. Chaturvedi*

SIGDIAL 2025

GRANTS AWARDED

Université de Toulouse

ANITI Graduate School Master Excellence Scholarship

2025–2026

TECHNICAL SKILLS

Languages: Python, C++, Bash

Systems & Performance: CUDA, Triton, Multi-GPU Training, Parallel Programming, Memory Optimization, Profiling (Nsight Systems, NVTX)

Algorithms & ML: PyTorch, Transformers, TensorFlow, Scikit-learn, Statistical Analysis

Infrastructure: Linux, Docker, Git, Slurm, CI/CD, AWS, GPU Clusters

Development: FastAPI, PyTest, Benchmarking

OPEN SOURCE CONTRIBUTIONS

pytorch/examples: Implemented Vision Transformer from scratch using only native PyTorch libraries, trained and evaluated on CIFAR-10 dataset

huggingface/optimum: Removed TF models support from optimum-onnx as it is not supported in Optimum v2

SELECTED PROJECTS

automanim

Nov 2025

FastAPI, Next.js, Manim, LLMs

LLM agent that generates and renders Manim animations from natural language, with tree-branching scene history for multi-turn edit/retry and pluggable model backends.

POSITIONS OF RESPONSIBILITY

Teaching Assistant/Mentor

Jun 2023–May 2025

ML Club, NIT Silchar

Conducted Python and Machine Learning sessions for freshmen and sophomores, facilitating peer learning and mentorship